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Investigation of synthesized carbazole derivative Cz-BPDB as a high-performance leveler for copper electroplating

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ABSTRACT

Electroplating additives, especially levelers of copper electroplating, are the most critical components of electroplating solutions in the manufacture of printed circuit boards and integrated circuits. A series of carbazole derivatives have been synthesized and used as levelers for through-hole electroplating. Electrochemical tests, theoretical calculations, and simulations were used to compare the various properties of the carbazole derivatives. Among them, the carbazole derivative Cz-BPDB was preferably subjected to electroplating which was demonstrated by metallographic microscopy, SEM, and XRD. To further investigate the mechanism of leveler Cz-BPDB, the nucleation model theory was used to explain the phenomenon of flat and smooth electroplating surfaces. The electrostatic potential plot and the average local ionization energy plot were employed to analyze the active sites of the Cz-BPDB. Moreover, the galvanostatic measurements were applied to investigate the interaction of the Cz-BPDB with PEG and SPS. Therefore, a more reasonable through-hole electroplating mechanism was proposed in electroplating solution with Cz-BPDB used as leveler. This work introduced an excellent performance leveler and contributed to the continuing development of the copper electroplating mechanism.

1. Introduction

With the wide application of 5G communication and related technologies, the miniaturization, lightweight, and high stability of various smart devices are increasingly valued and required. In this context, through-hole metallization in the manufacture of printed circuit boards (PCBs) [1,2] and the packaging of integrated circuits (ICs) [3–5] is one of the methods used to address the miniaturization of the devices. Copper, a low-resistance and high-conductivity metal, is also used as a through-hole metallization material for multilayer circuits to achieve faster transmission of electrical signals [6]. Therefore, to realize the metallization filling process of through holes, copper electroplating technology is proposed and developed continuously [7–9].

At present, the acidic copper sulfate solution is more commonly used for copper electroplating deposition [1], and the main components of the Virgin Make-up Solution (VMS) are copper sulfate ($CuSO_4$), sulfuric acid (H_2SO_4), and chloride ions (Cl^-) [10]. However, in through-hole electroplating, the orifice of through hole has higher current density and higher mass transfer [11]efficiency than the middle of through hole, resulting in very uneven metal deposition of through hole, which affects electrical signal transmission of the circuit and reduces the service life of electronic devices. Researchers have found that this problem can be solved by adding a variety of additives to the VMS, which has sparked increasing interest in the study of copper electroplating additives [12]. The additives of copper electroplating are generally divided into three categories: inhibitors (i.e. polyethylene glycol (PEG) [13–15], polypropylene glycol (PPG) [16] or their copolymers), accelerators (such as bis(3-sulfopropyl) disulfide (SPS) [17,18], or 3-mercapto-1-propanesulfonate (MPS) [19,20]), and levelers (i.e. Janus Green B (JGB) [11,21–23], Diazine Black [24], Alcian Blue, Safranin T [25], 2-mercaptopyridine [26,27], or 2,2-Dithiodipyridine [28]). With the synergistic effect of the three additives, a bright and uniform thickness of the electroplating inside and outside through hole is obtained. In previous work, our group reported aliphatic amine organics [29–31] and large conjugated molecules [32–39] as copper electroplating levelers, and all of these levelers performed excellently.

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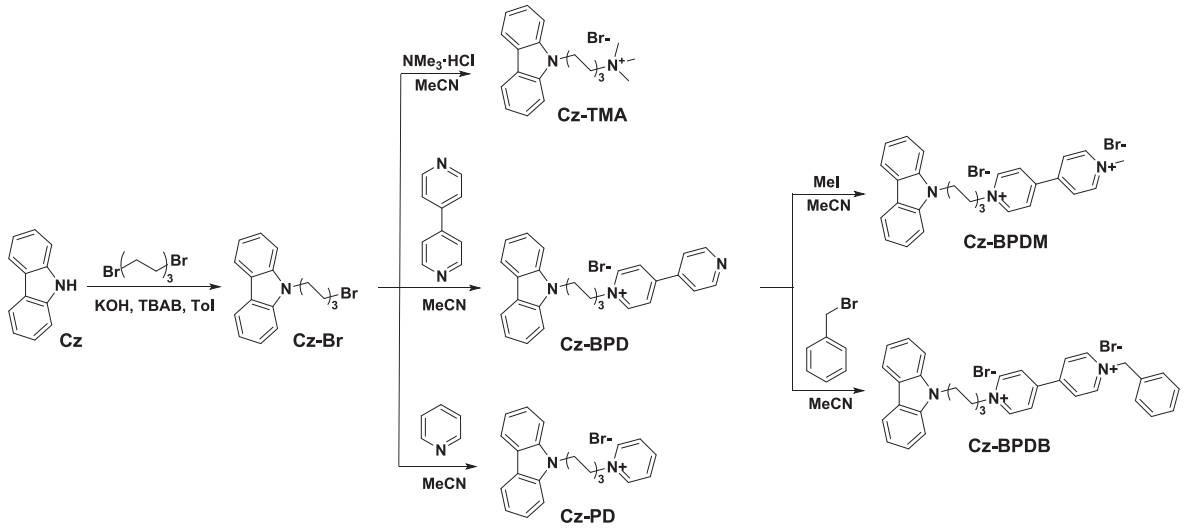


Fig. 1. The molecular structures and synthesis routes of five carbazole derivatives.

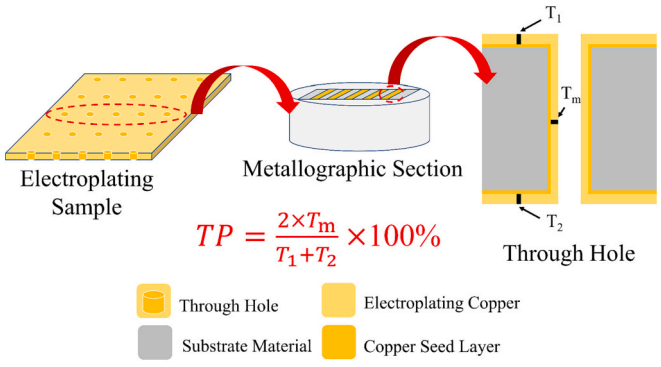


Fig. 2. The through hole diagram and the calculation formula of TP value.

In this work, a family of water-soluble carbazole derivatives were synthesized in high yield. Electrochemical tests were employed to explore the ability to inhibit the reduction and deposition of copper ions. In theoretical calculations, molecular orbital calculations of carbazole derivatives demonstrated the ability of carbazole derivatives to adsorb onto copper surfaces. Furthermore, molecular dynamics simulations visualized the adsorption behavior of carbazole derivatives on copper. After through-hole electroplating experiments, the effect of compound Cz-BPDB as a leveler was demonstrated by various instruments. Metallographic microscopy was applied to test throwing power (TP) values. SEM and XRD were respectively employed to observe the flatness of electroplating surface and the dominant crystal orientation of electroplating layer. In addition, to investigate the mechanism of through-hole electroplating, nucleation model theory, molecular properties, and electrochemical properties were further analyzed. This work increased the options for copper electroplating levelers with carbazole materials as a new plating leveler and contributed to the continuing development of the copper electroplating mechanism.

2. Material and methods

2.1. Materials synthesis and characterization

Five different compounds (Cz-TMA, Cz-PD, Cz-BPD, Cz-BPDM, and Cz-BPDB) were synthesized via two- or three-step reaction using commercially available carbazole as the starting substrate. The molecular structures and synthesis routes of the five carbazole derivatives are shown in Fig. 1. All the compounds obtained by synthesis were confirmed to be the target products by ^{1}H NMR,^{13}C NMR, and mass spectrometry. (The corresponding spectra and data for carbazole derivatives are shown in the Supporting Information.)

All reagents and chemicals in experiments were purchased from commercial sources and used without further purification. All reactions were monitored by TLC under UV light. ^{1}H NMR and ^{13}C NMR spectra were recorded by a Bruker AVANCE III 400 spectrometer operating at 400 MHz and 101 MHz, respectively. High-resolution mass spectra (HRMS) were obtained on a Xevo G2 TOF MS.

2.1.1. General procedure for the synthesis of compound Cz-Br

The reaction mixture of carbazole (0.1672 g, 1 mmol), 1,6-dibromohexane (1.2199 g, 5 mmol), potassium hydroxide (0.1683 g, 3 mmol), tetrabutylammonium bromide (0.3224 g, 1 mmol), and toluene (10 mL) was kept stirring for 3 h with air condensation in 35 °C. And then, the reaction solution was cooled to room temperature and stirred continuously for 12 h. When carbazole was completely consumed as indicated by TLC, the solvent was removed under reduced pressure. The crude product was purified by column chromatography using silica gel (petroleum ether and dichloromethane = 10:1–6:1, *v/v*) to give Cz-Br (0.2673 g, 81 % yield) as a white solid.

2.1.2. General procedure for the synthesis of compound Cz-TMA

The reaction mixture of Cz-Br (0.3303 g, 1 mmol), trimethylamine hydrochloride (0.4778 g, 5 mmol), sodium bicarbonate (0.42 g, 5 mmol), and acetonitrile (10 mL) was kept stirring for 12 h with Ar protection in 80 °C. After the reaction was finished, the reaction was cooled to room temperature. The solids in the reaction solution were filtered to obtain the filtrate. The crude product in the filtrate was purified by column chromatography using neutral alumina (dichloromethane and methanol = 5:1, *v/v*) to give Cz-TMA (0.3422 g, 88 % yield) as a white solid.

2.1.3. General procedure for the synthesis of compound Cz-PD

The reaction mixture of Cz-Br (0.3303 g, 1 mmol), pyridine (0.3955 g, 5 mmol), and acetonitrile (10 mL) was kept stirring for 24 h with Ar protection at 80 °C. After the reaction was finished, the reaction was cooled to room temperature. The reaction solution was poured into ethyl acetate (150 mL) for recrystallization to obtain the product Cz-PD (0.4038 g, 98 % yield) as a white solid.

2.1.4. General procedure for the synthesis of compound Cz-BPD

The reaction mixture of Cz-Br (0.3303 g, 1 mmol), 4,4′-bipyridine

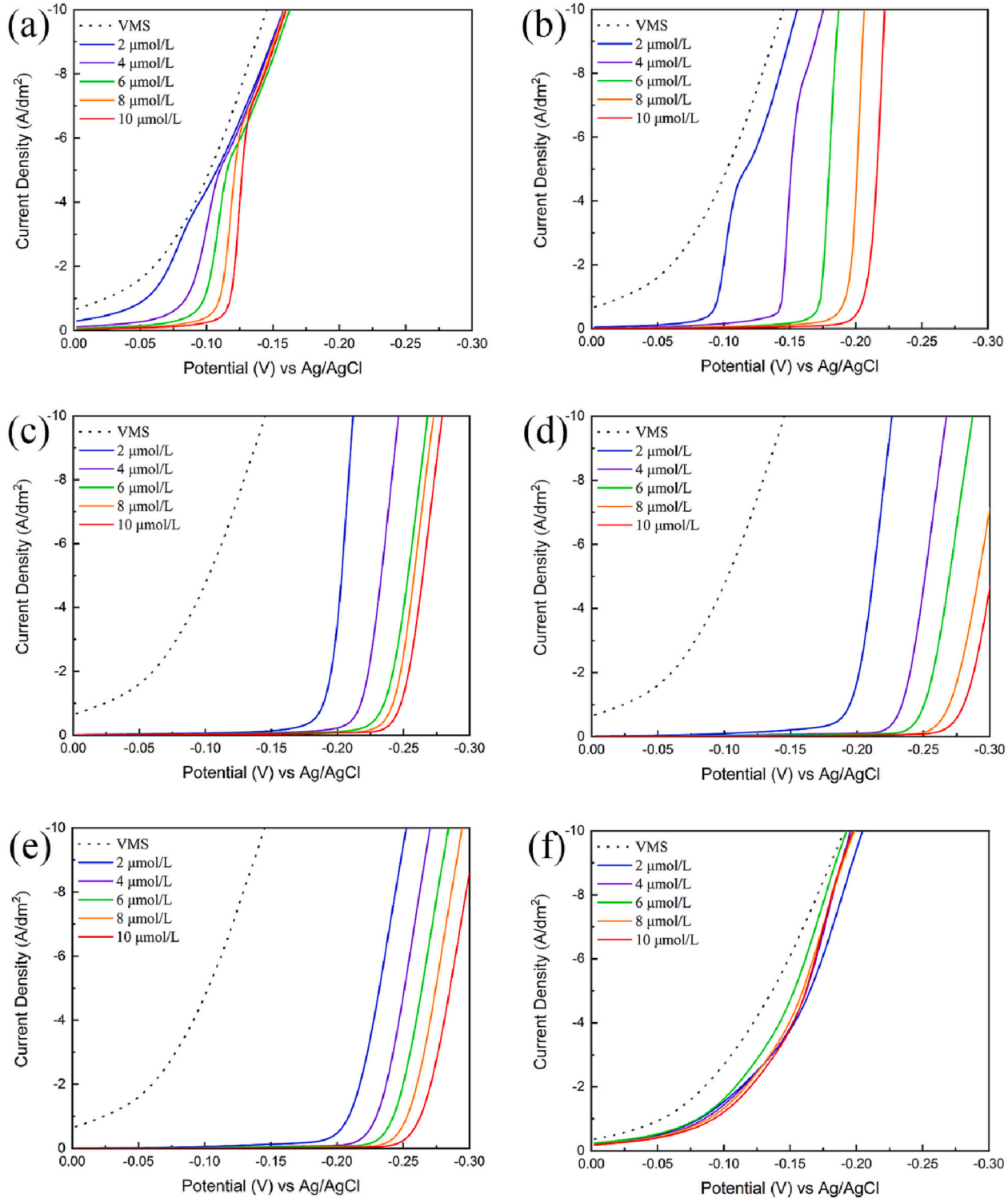


Fig. 3. Potentiodynamic polarization curves adding different concentrations of compounds in VMS. (a) Cz-TMA; (b) Cz -PD; (c) Cz -BPD; (d) Cz -BPDM; (e) Cz -BPDB; (f) Br^-.

(0.7809 g, 5 mmol), and acetonitrile (10 mL) was kept stirring for 12 h with Ar protection at 80 °C. After the reaction was finished, the reaction was cooled to room temperature. The reaction solution was poured into ethyl acetate (150 mL) for recrystallization to obtain the product Cz-BPD (0.4326 g, 89 % yield) as a yellow solid.

2.1.5. General procedure for the synthesis of compound Cz-BPDM

The reaction mixture of Cz-BPD (0.4865 g, 1 mmol), iodomethane (0.7097 g, 5 mmol), and acetonitrile (20 mL) was kept stirring for 12 h with Ar protection at 80 °C. After the reaction was finished, the reaction was cooled to room temperature. The reaction solution was poured into ethyl acetate (150 mL) for recrystallization to obtain the product Cz-BPDM (0.5682 g, 98 % yield) as a yellow solid.

2.1.6. General procedure for the synthesis of compound Cz-BPDB

The reaction mixture of Cz-BPD (0.4865 g, 1 mmol), (bromomethyl)

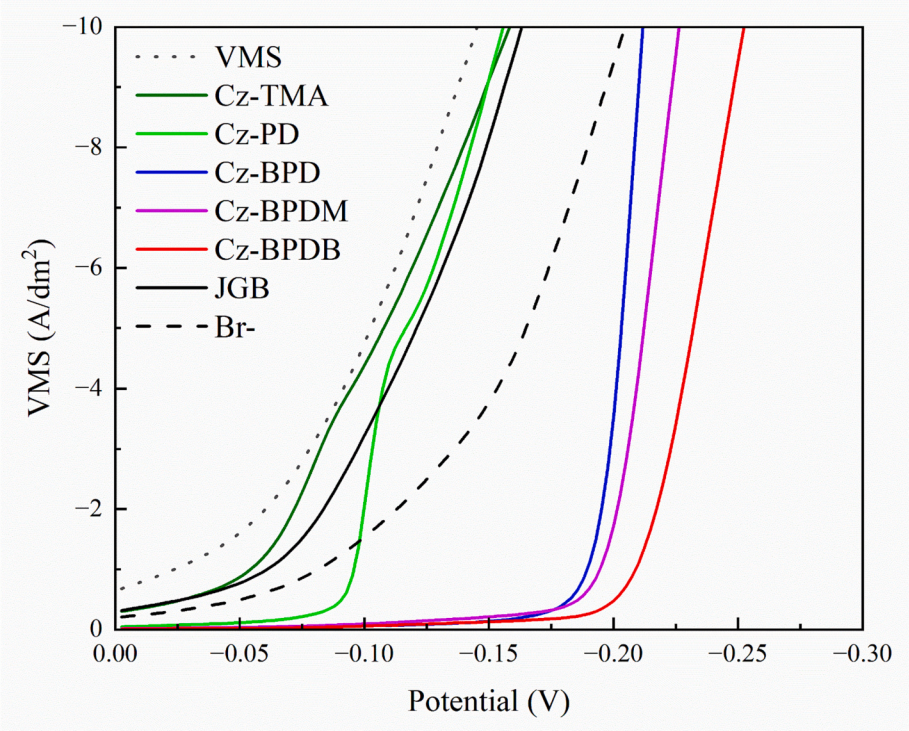


Fig. 4. Potentiodynamic polarization curves adding 2 μmol/L compounds.

benzene (0.8552 g, 5 mmol), and acetonitrile (20 mL) was kept stirring for 12 h with Ar protection in 80 °C. After the reaction was finished, the reaction was cooled to room temperature. The reaction solution was poured into ethyl acetate (150 mL) for recrystallization to obtain the product Cz-BPDB (0.5415 g, 82 % yield) as a red solid.

2.2. Electrochemical measurements

All electrochemical measurements were performed by an electrochemical workstation PGSTAT302N Auto-Lab with a standard three-electrode system. The working electrode (WE) was a platinum rotating disk electrode (Pt-RDE) with a 5 mm. Moreover, a Pt stick with a diameter of 2 mm was serve as the counter electrode (CE) and the reference electrode (RE) was a silver/silver chloride (Ag/AgCl) electrode. The VMS included 60 g/L $CuSO_4$, 100 mL/L H_2SO_4, and 140 μL/L HCl in all electrochemical tests. In addition, polyethyleneglycol (PEG, MW = 10,000) and bis-(sodium-sulfopropyl)-disulfide (SPS) respectively played the role of inhibitor and accelerator.

Before conducting each potentiodynamic polarization test or linear sweep voltammetry (LSV) measurement, Pt-RDE, which was pre-treated for 120 s to form a thin copper layer, was transformed into Cu-RDE. Furthermore, a constant scan rate was 20 mV/s with a scanning potential range of 0 to −0.8 V in all potentiodynamic polarization tests. The scanning potential ranged from 1.57 to −0.20 V under a constant scan rate of 50 mV/s for all cyclic voltammetry (CV) measurements in this essay. In galvanostatic measurements (GMs), a current density was fixed at 1.5 A/dm^2. In addition, a glassy carbon rotating disk electrode (GC-RDE, Φ 5 mm) was used as WE with a copper sheet as CE and an Ag/AgCl electrode as RE for all the chronoamperometry (CA) tests.

2.3. Calculational methods and details of simulations

Quantum chemical calculations were conducted on Gaussian 09 program package. The geometry optimizations of all carbazole derivatives were based on density functional theory (DFT) using the B3LYP method with a 6-311G+ basis set. To obtain more precise data on the highest occupied molecular orbital (HOMO) energy (E_{HOMO}) and the lowest unoccupied molecular orbital (LUMO) energy (E_{LUMO}), the energy calculations were carried out under B3LYP/6-311G+ (d, p) level. ΔE was defined as $\Delta E = E_{LUMO}$ - E_{HOMO}. The molecular orbital, the electrostatic potential (ESP), and the average local ionization energy (ALIE) were calculated, analyzed, and drawn by using Multiwfn [40–43] and VMD program package.

The molecular dynamics (MD) simulations were performed by the Forcite module in Materials Studio (from Accelrys Inc). The adsorption interaction between each carbazole derivative on Cu (111) surface was studied under the environment of 298 K, NVT ensemble, and COMPASS II force field with a step time of 1 fs and simulation time of 500 ps. The MD simulations were shown in simulation boxes (4.10 × 4.10 × 7.10 nm^3) including a carbazole derivative molecule, 1000 water molecules, and Cu (111) surface. The adsorption energy of a carbazole derivative with the Cu (111) surface was defined by the following Eqs. (1) and (2).

$$E_{ads} = E_{complex} - (E_{Cu} + E_{additive}) \quad (1)$$

$$E_{binding} = -E_{ads} \quad (2)$$

where $E_{complex}$ represented the total energy of the simulated system, E_{Cu} and $E_{additive}$ referred to the energy of the copper crystalline surface and the energy of only organic molecules, respectively. A larger value of $E_{binding}$ indicated that a molecule was easily adsorbed to the Cu (111) surface.

2.4. Electroplating experiments

A double-sided copper-clad board with many through holes was used as an electroplating sample. The size of the electroplating board was 15 × 5 cm^2. Moreover, the diameter and depth of through holes on the electroplating board were 300 μm and 3 mm, respectively. Before electroplating experiments, an electroplating board underwent a pre-treatment process containing degreased by a degreasing agent, washed with deionized water, and acid washed with 10 % H_2SO_4 solution. The middle of a 1500 mL Harring cell was placed into an electroplating

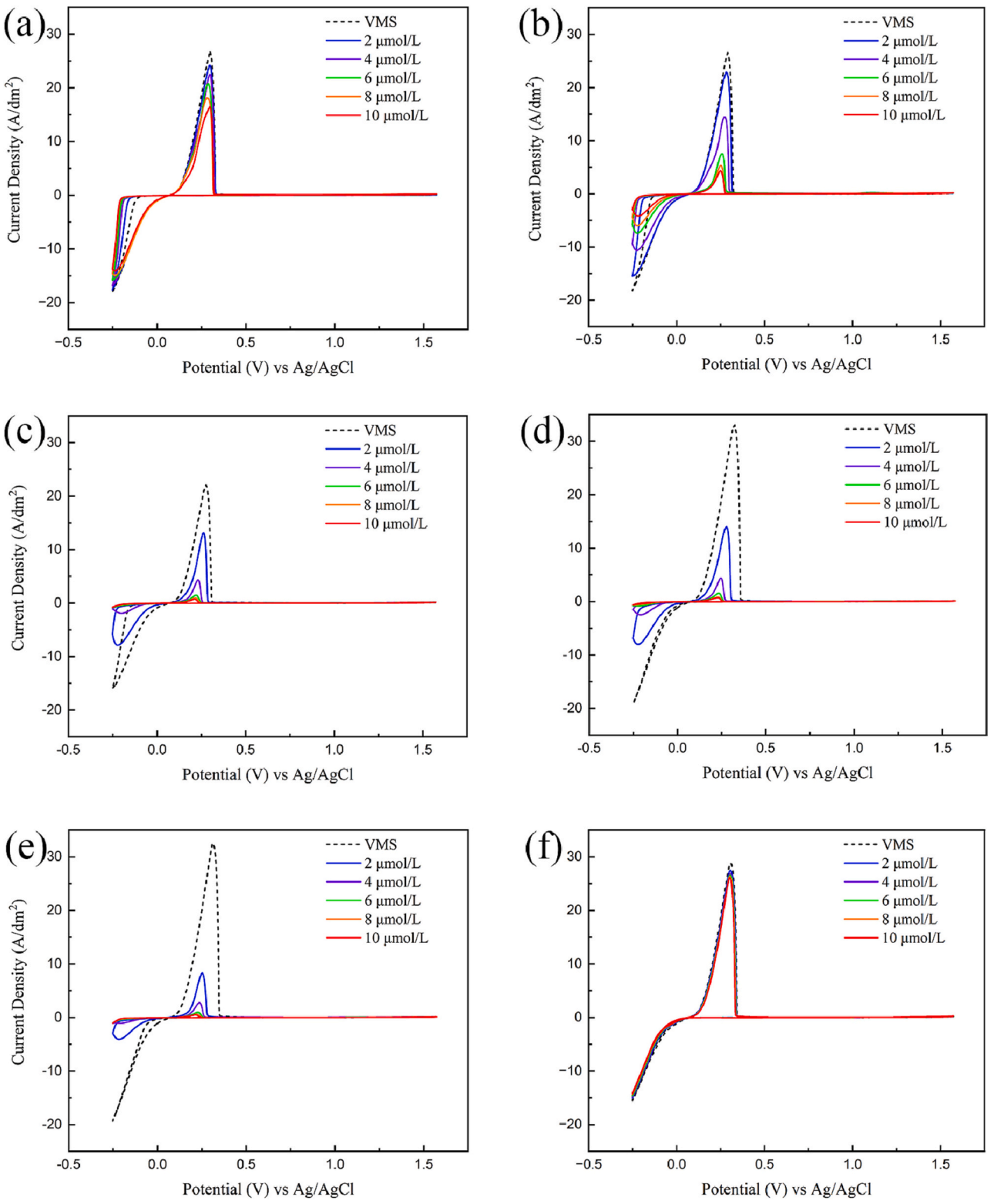


Fig. 5. Cyclic voltammetric curves adding different concentrations of compounds in VMS. (a) Cz-TMA; (b) Cz -PD; (c) Cz -BPD; (d) Cz -BPDM; (e) Cz -BPDB; (f) Br^-.

board as the cathode, and two phosphorus-containing Cu plates were placed on both sides of the cell as the anode. During electroplating, a continuous flow of air bubbles was kept for enduring good mass transfer under 1.5 A/dm^2 current density and 75 min electroplating time. The electroplating solution formula was based on the VMS with the addition of inhibitor (PEG, MW = 10,000), accelerator (SPS), and leveler (Cz-BPDB or JGB).

2.5. Physical properties of copper deposition

Metallographic microscopy was performed to show the thickness of copper electroplating deposition in the middle and outside of through holes to calculate the TP values of different formulas. The calculation method of the TP values is shown in Fig. 2. In addition, scanning electron microscopy (SEM, Sigma 300) was employed to observe the roughness of

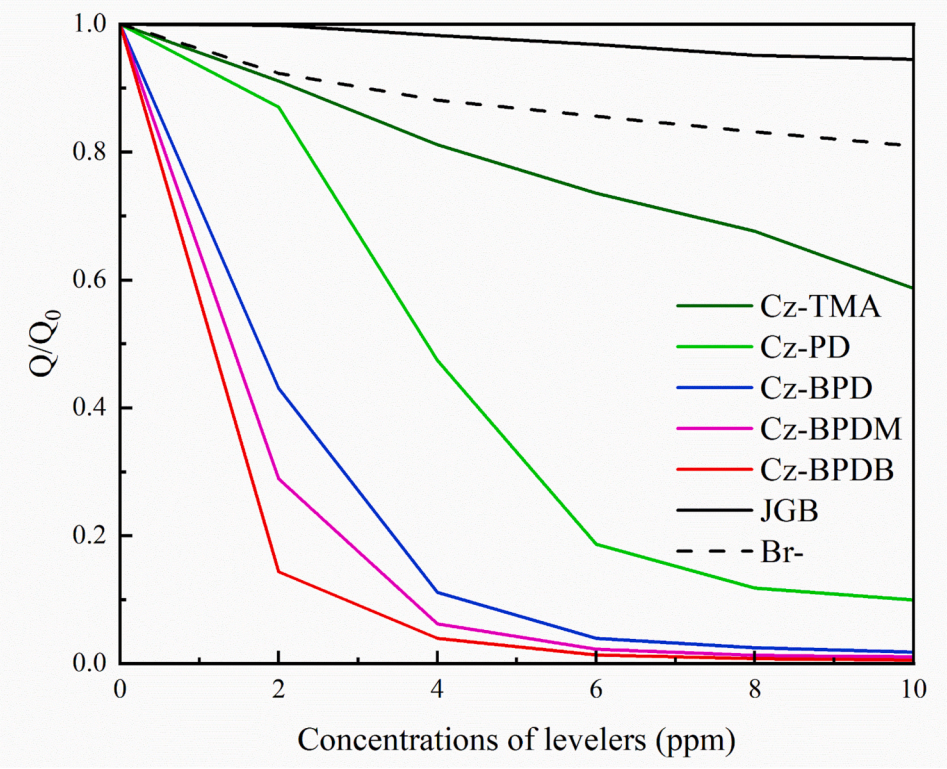


Fig. 6. The ratio (Q/Q_0) plot of copper stripping peak areas.

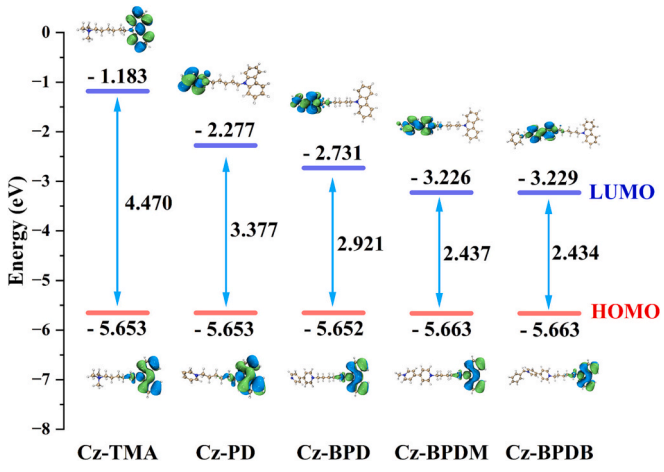


Fig. 7. Distributions and orbital energy values of HOMO orbital and LUMO orbital of carbazole derivatives.

electroplating surfaces from various electroplating solutions. X-ray diffraction (XRD, Bruker D2 Phaser) was used to reveal the preferential orientation of electroplating surfaces for various electroplating solutions.

3. Results and discussion

3.1. Potentiodynamic polarization measurements

Potentiodynamic polarization curves were utilized to assess the ability of electroplating additives to inhibit the reduction and deposition of copper ions in electroplating solutions [32,44]. When the current density began to grow during the scanning potential test from positive to negative scanning, copper ions began to be reduced to copper and deposited on the cathode. The potential at which the current density began to increase was called the deposition potential. After adding organics as additives to the VMS, the higher deposition potential indicated that the organics were more capable of inhibiting the reduction and deposition of copper ions. In addition, a study [45] found that Br^- had a positive effect on electroplating. In our studies, a series of comparative tests were done with VMS solutions containing the same concentration of bromine ions (Br^-) as a comparison sample.

The five carbazole derivatives and Br^- were characterized at different concentrations as shown in Fig. 3. The commercial JGB was characterized at different concentrations as shown in Fig. S1 in the Supporting Information. In the VMS without any additives, the current density increased as the scanning potential scanned from positive to negative from 0 V. And as can be seen in Fig. 3(a)-(e), the deposition potential increased as the concentration of carbazole derivatives increased. To exclude the effect of Br^- on potentiodynamic polarization tests, a VMS solution containing Br^- was tested and the results were shown in Fig. 3(f). The deposition potential of the VMS solution containing Br^- was slightly larger than that of the VMS solution and much smaller than that of the carbazole derivatives (Cz-PD, Cz-BPD, Cz-BPDM, and Cz-BPDB), which implied that Br^- had a weak effect on the deposition potential. Fig. 4 shows the potentiodynamic polarization curves of adding 2 μmol/L of various compounds to VMS, which showed the order of the deposition potential (i.e. VMS < Cz-TMA < Cz-PD < Cz-BPD < Cz-BPDM < Cz-BPDB). As a result, it was obvious that Cz-BPDB was the strongest of the five carbazole derivatives in inhibiting the deposition of copper ions by reduction, and its inhibition ability was greater than that of commercial JGB, which had great potential as a better leveler for copper electroplating.

3.2. Cyclic voltammetry measurements

In electrochemical measurements, cyclic voltammetric curves were performed to test the ability of organics to inhibit the reduction and deposition of copper ions in the electroplating solution. The negative

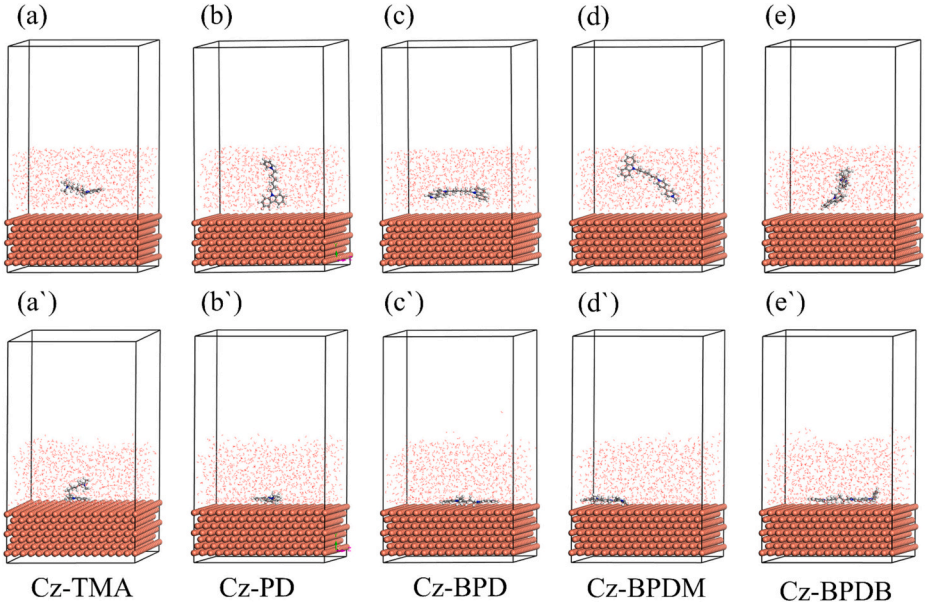


Fig. 8. The pre-simulation (a) - (e) and post-simulation (a`) - (e`) states of molecular dynamics simulation of five carbazole derivatives on Cu (111) surfaces.

Table 1
The binding energies of five carbazole derivatives on Cu (111) surfaces.

Organ	Cz-TMA	Cz-PD	Cz-BPD	Cz-BPDM	Cz-BPDB
$E_{complex}$	−52,981.87	−53,055.60	−52,954.14	−52,911.54	−52,916.05
E_{Cu}	−53,145.94	−53,145.79	−53,145.63	−53,145.78	−53,145.68
$E_{additive}$	220.98	188.77	305.34	351.10	363.56
E_{ads}	−56.91	−98.58	−113.85	−116.86	−133.93
$E_{binding}$	56.91	98.58	113.85	116.86	133.93

Table 2
The formulas of through-hole electroplating baths.

	SPS (ppm)	PEG (ppm)	Leveler	Leveler (ppm)
Bath 1	0	0	–	0
Bath 2	2	200	–	0
Bath 3	2	200	Cz-BPDB	0.1
Bath 4	2	200	Cz-BPDB	0.3
Bath 5	2	200	Cz-BPDB	0.5
Bath 6	2	200	JGB	0.1
Bath 7	2	200	JGB	0.3
Bath 8	2	200	JGB	0.5

current density peak in the cyclic voltammetric curve was the copper deposition peak, which represented the ability of copper ions to be reduced to copper and deposited. While, the positive current density peak was the copper stripping peak, whose area represented the ability of the deposited copper to be oxidized to copper ions. The peak area of the copper stripping peak was positively correlated with the peak area of the copper deposition peak. Therefore, the smaller the peak of the copper deposition peak, the smaller the peak area of the copper stripping peak, which implied the stronger ability of the organics to inhibit the reduction and deposition of copper ions [29,46].

The cyclic voltammetric curves of the five carbazole derivatives and Br^- were shown in Fig. 5. The cyclic voltammetric curves of the commercial JGB were shown in Fig. S2. As seen in Fig. 5(a)-(e), the copper stripping peak of the VMS without any organic additives had the largest peak area, and the peak area of the copper stripping peak decreased as the concentration of the additives added to the electroplating solution increased. This indicated the carbazole derivatives had the ability to inhibit the reduction and deposition of copper ions.

To exclude the instability of electrochemical workstation instrumentation on electrochemical tests, the ratio of the peak area of the copper stripping peak (Q) of the electroplating solution containing different concentrations of additives to the peak area of the copper stripping peak (Q_0) of the electroplating solution without any additives was proposed and adopted [47]. And smaller value of Q/Q_0 represented greater ability of the organic additive to inhibit the reduction and deposition of copper ions in the electroplating solution. As shown in Fig. 6, five carbazole derivatives all inhibited the ability of copper ions to be reduced and deposited in this order of magnitude (i.e. Cz-TMA < Cz-PD < Cz-BPD < Cz-BPDM < Cz-BPDB) and in agreement with the conclusion of the polarization curves. As shown in Figs. 5 (f) and 6, the Q/Q_0 value of VMS containing Br^- was greater than those of the synthesized carbazole derivatives, which implied a weak effect of Br^- on the ability to inhibit the reduction and deposition of copper ions. Meanwhile, all carbazole derivatives were stronger than the commercial JGB in inhibiting the reduction and deposition of copper ions. Among them, Cz-BPDB had the strongest inhibitory ability and had great potential to be a higher-quality leveler for copper electroplating.

3.3. Quantum chemical calculations

Quantum chemical calculations were used to evaluate the potential of organics as electroplating levelers, and the results were shown in Fig. 7. According to the front-line orbital theory, E_{HOMO} and E_{LUMO} were correlated with the electron-giving and electron-absorbing ability of the molecule, respectively. Higher E_{HOMO} values represented greater tendency of the molecule to give electrons, while lower E_{LUMO} values meant greater tendency of the molecule to give trapped electrons. Based on previous studies, a smaller value of ΔE represented stronger adsorption of organic molecules on copper [48,49].

As shown in Fig. 7, the HOMO orbitals of carbazole derivatives were

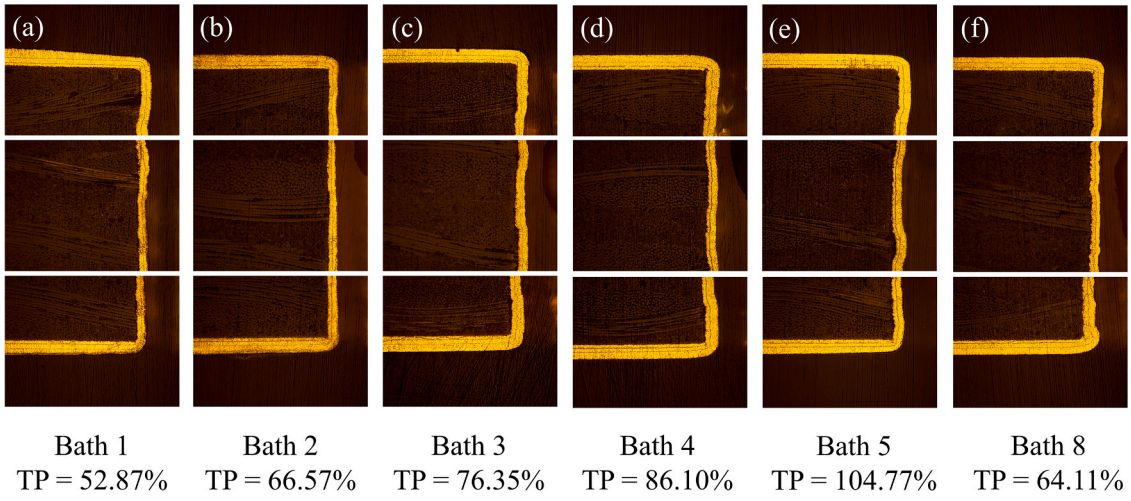


Fig. 9. The cross-sectional images of through holes obtained from Bath1–5 and Bath 8 after copper electroplating experiments.

Table 3
Through-hole measurement data and TP values.

	T_1/μm	T_m/μm	T_2/μm	TP, mean ± SD/%
Bath 1	22.59	10.7	18.83	52.87 ± 1.79
Bath 2	20.76	14.34	23.57	66.57 ± 2.20
Bath 3	20.53	16.05	22.47	76.35 ± 1.78
Bath 4	22.05	18.19	21.42	86.10 ± 2.28
Bath 5	20.76	20.96	20.33	104.77 ± 2.39
Bath 8	19.27	13.14	21.41	64.11 ± 0.84

distributed on the carbazole conjugated structure and their E_{HOMO} values were close to each other, which indicated the similar electron-giving ability of the carbazole derivatives. However, the E_{LUMO} values of the five carbazole derivatives differed considerably, which indicated that the electron-capturing ability of the carbazole derivatives was different, with Cz-BPDB having the greatest electron-capturing ability. In addition, the ΔE values exhibited an order (i.e. Cz-TMA < Cz-PD < Cz-BPD < Cz-BPDM < Cz-BPDB) which was consistent with the electrochemical results. Cz-BPDB had the smallest ΔE value, which implied stronger adsorption on copper.

3.4. Molecular dynamic simulations

To visualize the adsorption behavior of carbazole derivatives on copper, molecular dynamics simulations were employed to carry out further studies [50,51]. The pre-simulation and post-simulation states of the carbazole derivatives on the Cu (111) surface were shown in Fig. 8. The pre-simulation and post-simulation states of JGB molecules on the Cu (111) surface were shown in Fig. S3. Table 1 showed the adsorption energies of carbazole molecules on the Cu (111) surface. As shown in Fig. 8, carbazole derivatives were not on the copper surface before the simulation; after 500 ps time of simulation, all carbazole derivatives were adsorbed to the copper surface, which was in combination with the results of electrochemical tests, proved that the synthesized carbazole derivatives were adsorbed to the copper surface to prevent the reduction and deposition of copper. While the specific adsorption energy values were given in Table 1, Cz-BPDB had the largest adsorption energy of 133.93 kcal/mol. And the same order of magnitude of adsorption energy was found in electrochemical tests and theoretical calculations (i.e. Cz-TMA < Cz-PD < Cz-BPD < Cz-BPDM < Cz-BPDB).

3.5. Copper electroplating experiments and characterization

Based on the above results, Cz-BPDB exhibited optimal performance in the electrochemical tests, calculations, and simulations and was selected as the copper electroplating leveler to be added to the electroplating solution for copper electroplating experiments. Therefore, based on the VMS, various copper electroplating formulas were designed as shown in Table 2, and these formulas were used for copper electroplating experiments in order to characterize the through-hole filling properties (i.e. TP value), the surface morphology and the dominant crystal orientation of the electroplating surface.

After copper electroplating experiments, the cross-sectional images of through holes in Bath 1–5 were taken by metallographic microscope, as shown in Fig. 9 (a)-(e). The through-hole filling performance of JGB as a leveler was shown in Fig. 9 (f). The data from the through-hole copper electroplating experiments were shown in Table 3. According to previous studies, a TP value greater than 100 % implied that the through hole had the same thickness of copper deposited inside and outside the through hole, which was an important indicator of the electroplating filling of the through hole [39,44].

In Bath 1 electroplating solution without any additives, the TP value of through-hole electroplating filling was 52.87 %, which represented that the thickness of the electroplating in the middle of through hole was only half of which in the through-hole orifice, i.e., the filling effect was poor. In Bath 2 electroplating solution obtained by adding 200 ppm PEG and 2 ppm SPS to Bath 1 electroplating solution, the TP value increased to 66.57 %, but this still did not reach the desired requirement. In Bath 3, Bath 4, and Bath 5 electroplating solutions with 0.1 ppm, 0.3 ppm, and 0.5 ppm Cz-BPDB, respectively, the electroplating layers were dramatically improved. Among them, the TP value of Bath 5 was 104.77 %, which was in accordance with the requirement of through-hole electroplating. It can be concluded that the addition of Cz-BPDB to the copper electroplating solution increased the thickness of the electroplating layer in the middle of the through-hole. At the same time, the TP value of Bath 8 including 0.5 ppm JGB was only 64.11 %, which was far less effective than Cz-BPDB at the same concentration. Moreover, as shown in Fig. S4(a), the TP value of Bath 9 containing Br^- was 58.94 %, which shows the ineffectiveness of Br^- in the improvement of through-hole electroplating.

Subsequently, the effect of Cz-BPDB as a copper electroplating leveler on the surface morphology of the electroplating layer was further investigated by SEM [34,52]. As shown in Fig. 10, the surface of the electroplating layer obtained by electroplating deposition under Bath 1 electroplating solution without any additives was particularly rough and uneven with the presence of grooves. In Bath 2 electroplating solution with 200 ppm PEG and 2 ppm SPS, the electroplating surface was improved, but it was still obviously rough and uneven. In Bath 3, Bath 4, and Bath 5 electroplating solutions with 0.1 ppm, 0.3 ppm, and 0.5 ppm

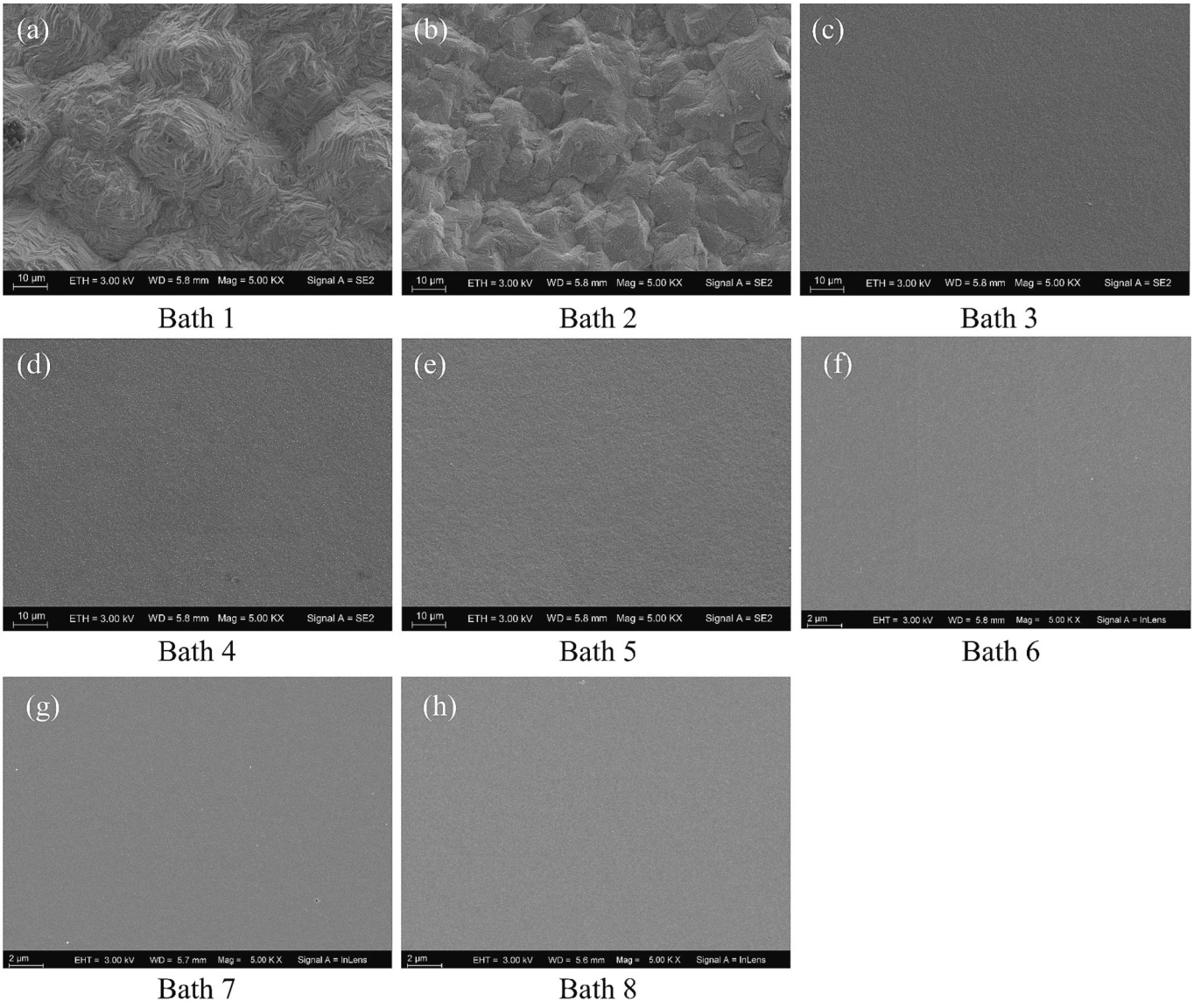


Fig. 10. SEM images of copper electroplating layer in Bath 1–8.

Cz-BPDB, respectively, the electroplating surface was greatly improved and became smooth and flat. It can be seen that the addition of Cz-BPDB to the copper electroplating solution could make the electroplating layer smooth and flat. In addition, as shown in Fig. 10 (f)-(h), the improvement of the surface morphology of the electroplating layer by Cz-BPDB was close to that of JGB.

Meanwhile, in order to investigate the effect of Cz-BPDB as a leveler on the dominant crystal orientation of the electroplating layer, XRD was employed to test the crystallographic orientation of the electroplating layer [39,53], and the test results were shown in Fig. 11. In Bath 1 electroplating solution, the dominant crystal orientation of the electroplating layer was (220) crystalline surface accounted for 100 %. In Bath 2 electroplating solution in the presence of PEG and SPS, the crystalline tendency of the electroplating layer was changed by adding (200) and (111) crystalline surface and the preferential orientation became (200). After adding Cz-BPDB to the electroplating solution, the crystalline tendency of the electroplating layer changed sharply, the preferential orientation became (111), and the percentage of (111) crystalline surface increased with the increase of Cz-BPDB concentration. However, the preferential orientation of JGB is (200) under the same concentration condition as shown in Fig. 11. It can be seen that the addition of Cz-BPDB changed the preferential orientation of the electroplating layer toward the lower energy and more stable Cu (111) crystalline surface.

In short, Cz-BPDB was superior to JGB in TP value and XRD preferential orientation, and Cz-BPDB was close to JGB in the flatness of the electroplating layer, which showed that Cz-BPDB was an excellent performing copper electroplating leveler.

3.6. *Exploration of the mechanism of electroplating additives*

In the presence and absence of Cz-BPDB in the VMS containing PEG and SPS (i.e., VPS + Cz-BPDB and VPS), the results of CA tests for VPS + Cz-BPDB and VPS were made out at different potentials as shown in Figs. 12. and S6, respectively. Before the CA tests, the LSV tests were performed to obtain the copper deposition potential as shown in Fig. S5. Because the potentials for CA tests needed to be less than the copper deposition potential in the VPS + Cz-BPDB and VPS solutions and greater than the hydrogen precipitation potential, these potentials (i.e., −0.24, −0.26, −0.28, and −0.30 V) were screened for CA tests.

Information on the nucleation and nuclei growth of copper deposition was obtained by CA tests. The Scharifker-Hills model gave a three-dimensional nucleation theory of progressive nucleation and instantaneous nucleation [54–56]. The number of nuclei did not change with time (i.e., no new nuclei were generated during nuclei growth) was

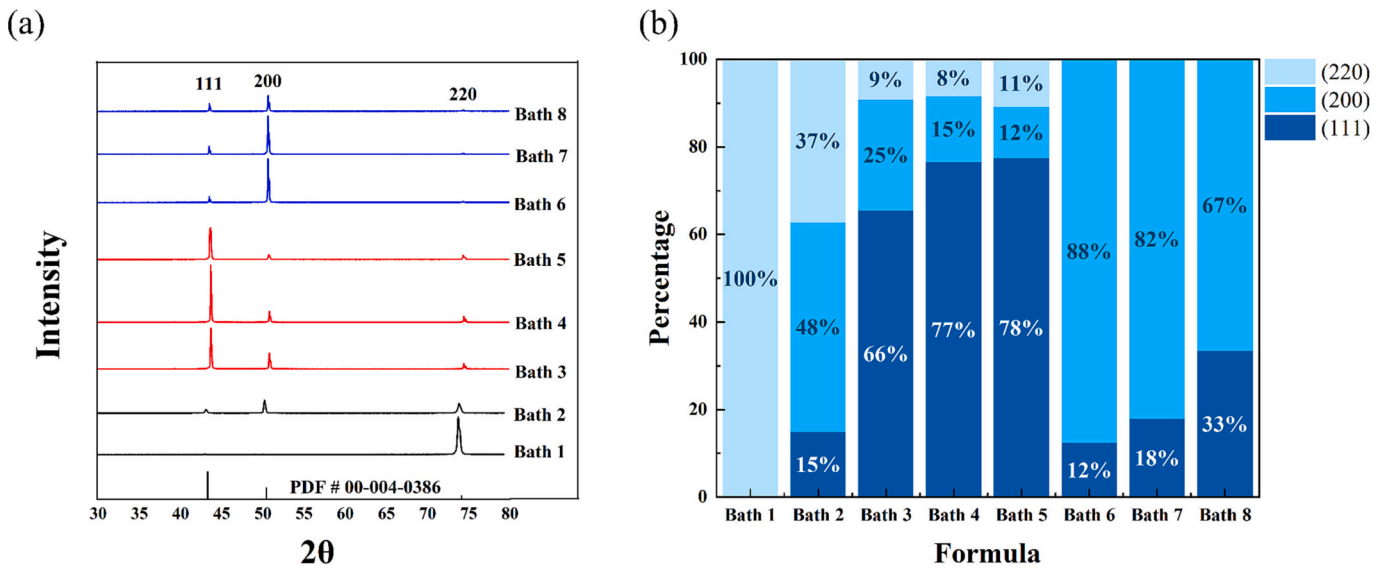


Fig. 11. XRD patterns of copper electroplating layer in Bath 1–8.

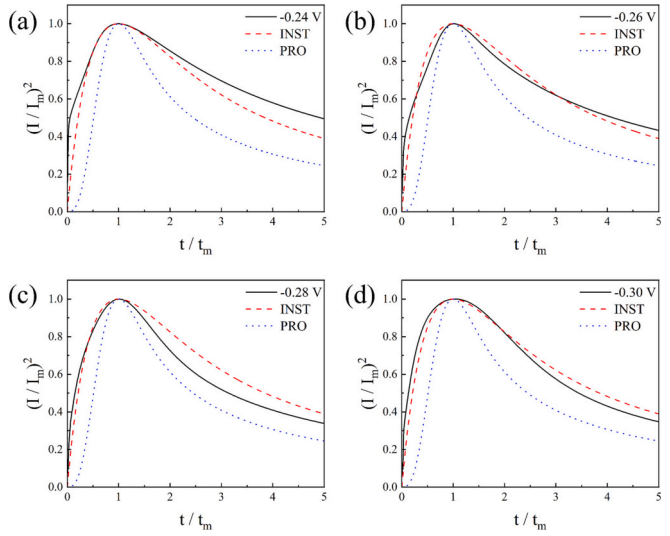


Fig. 12. The 3D instantaneous and progressive nucleation plots of Scharifker and Hills model, and the non-dimensional plots (symbol) of VPS + Cz-BPDB.

called the instantaneous nucleation process. While the number of nuclei changed with time and was a function of time (i.e., new nuclei were generated during nucleus growth), which was called the progressive nucleation process. The transient current and time in the nucleation theory were analyzed as shown in Table 4. Here, I was the current density and t denotes time; I_m denotes the maximum current density and t_m was the time corresponding to I_m.

To investigate the mechanism of action of the electroplating leveler, CA tests were employed to analyze the effect of Cz-BPDB on the reduction of copper ions to obtain copper nucleation. As shown in

Table 5
Best-fit parameters and kinetic data resulting from analysis of the experimental current–time transient.

System	Potential (V)	t_m(s)	-I_m (A·cm^{-2})	10^{-14} D, mean ± SD (cm^2·s^{-1})	10^{10} N, mean ± SD (cm^{-2})
VPS	−0.24	0.73	37.88 × 10^{-3}	18.87 ± 0.76	19.97 ± 2.92
	−0.26	0.56	48.53 × 10^{-3}	25.20 ± 0.75	21.03 ± 1.55
	−0.28	0.43	61.97 × 10^{-3}	31.27 ± 0.81	23.26 ± 0.15
	−0.30	0.35	74.47 × 10^{-3}	37.20 ± 1.37	24.90 ± 0.43
VPS + Cz-BPDB	−0.24	1.66	27.84 × 10^{-3}	23.90 ± 0.36	7.87 ± 0.59
	−0.26	0.92	39.06 × 10^{-3}	28.20 ± 2.57	12.43 ± 1.29
	−0.28	0.63	54.50 × 10^{-3}	35.87 ± 1.27	14.40 ± 0.78
	−0.30	0.4	69.25 × 10^{-3}	38.40 ± 2.78	20.37 ± 1.82

Table 4
The instantaneous and progressive nucleation of expressions resulting from the analysis of the current maxima.

Instantaneous nucleation	Progressive nucleation
$\left(\frac{I}{I_m}\right)^2 = \frac{1.9542}{\frac{t}{t_m}}\left\{1 - exp\left[-1.2564\left(\frac{t}{t_m}\right)\right]\right\}^2$	$\left(\frac{I}{I_m}\right)^2 = \frac{1.2254}{\frac{t}{t_m}}\left\{1 - exp\left[-2.3367\left(\frac{t}{t_m}\right)^2\right]\right\}^2$
$t_m = \frac{1.2564}{N\pi kD}$	$t_m = \left(\frac{4.6733}{AN_\infty \pi k' D}\right)\frac{1}{2}$
$I_m = 0.6382zFDc(kN)^{\frac{1}{2}}$	$I_m = 0.4615zFD^{\frac{3}{4}}c(k'AN_\infty)^{\frac{1}{4}}$
$I_m^2 t_m = 0.1629(zFc)^2 D$	$I_m^2 t_m = 0.2598(zFc)^2 D$
$k = \left(\frac{8c\pi M}{\rho}\right)\frac{1}{2}$	$k = \frac{4}{3}\left(\frac{8c\pi M}{\rho}\right)\frac{1}{2}$

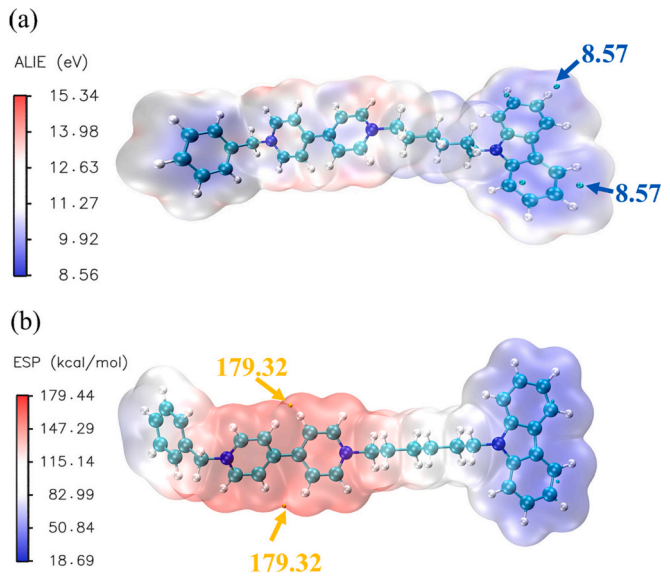


Fig. 13. (a) The ESP mapped molecular van der Waals surface of Cz-BPDB. (b) The ALIE mapped molecular van der Waals surface of Cz-BPDB.

Fig. 12, the red line was the instantaneous nucleation model and the blue line was the progressive nucleation model. In Fig. S6(a)-(d), the VPS electroplating system deviated from the nucleation model but converged to the instantaneous nucleation model in the VPS electroplating solution without the presence of Cz-BPDB. As in Fig. 12(a)-(d), the VPS + Cz-BPDB electroplating system further converged to the instantaneous nucleation model in the VPS + Cz-BPDB electroplating solution with the addition of 0.5 ppm Cz-BPDB. Among them, the instantaneous nucleation model was fitted well at the deposition potentials of −0.26 and −0.30 V. As shown in Table 5, *D* denoted the ion diffusion coefficient and *N* was the number of copper nuclei per square centimeter. After adding 0.5 ppm Cz-BPDB to the electroplating solution, *D* increased while *N* decreased, which indicated that Cz-BPDB inhibited the formation of copper nuclei (i.e., inhibited the reductive deposition of copper ions). Meanwhile, this was consistent with the polarization curves and cyclic voltammetry tests. In conclusion, it can be drawn that Cz-BPDB could adsorb to the electroplating surface and prevent the reduction and deposition of copper ions.

In the conclusion of above CA tests that Cz-BPDB could adsorb to copper surfaces, the ALIE plot and the ESP plot, the powerful molecular visualization tools, were used to investigate the molecular structural properties and preferential adsorption sites. The ALIE plot and the ESP plot were calculated and defined by the following Eqs. (3) and (4), respectively.

$$\bar{I} = \frac{\sum_{i\in occ} \rho_i(r)|\varepsilon_i|}{\rho(r)} \tag{3}$$

$$V_{tot}(r) = V_{nuc}(r) + V_{ele}(r) = \sum_A \frac{Z_A}{|r - R_A|} - \int \frac{\rho(r')}{|r - r'|} dr' \tag{4}$$

The ALIE plot described the reactivity of the electrons in the molecule, where lower values of the average ionization energy indicated weaker binding and higher electron activity [39,50]. As shown in Fig. 13 (a), the low value of the average ionization energy (8.57 eV) was mainly distributed on the carbazole ring of Cz-BPDB, which indicated the high activity of the electrons on the carbazole ring of Cz-BPDB and the easy chemisorption to the electron-rich copper surface for electrophilic reaction. In addition, the ESP plot demonstrated the distribution of electron density, where high electrostatic potential values represent low current density [50,51]. As shown in Fig. 13(b), the high value (179.32 kcal/mol) in the ESP plot was mainly distributed on the positively charged bipyridine conjugate ring of Cz-BPDB, which indicated that the current density here was the lowest in the whole molecule. During electroplating, the positively charged bipyridine conjugate ring of Cz-BPDB tended to physically adsorb to the electron-rich copper surface to prevent the reduction and deposition of the copper ions.

To discuss in further detail the mechanism of additives on the filling of through-hole electroplating, GMs tests were used for solutions with different addition sequences or different rotational speeds [31,57], and the results of these tests were shown in Fig. 14. As shown in Fig. 14(a), at 1000 s, 200 ppm PEG was added to prevent copper ions from reductive deposition. This caused a sharp drop in potential (−0.04 to −0.25 V) and polarization. SPS promoted the reduction and deposition of copper ions after being added to the 1000 rmp-PSC bath at 2000 s; as a result, the potential significantly rose (from −0.25 to −0.15 V), and depolarization took place. When Cz-BPDB was added to the 1000 rmp-PCS solution, it also prevented copper ions from being reduced and deposited. A polarization and a modest drop in potential (−0.25 to −0.26 V) were noticed. These showed, respectively, PEG's antagonistic and synergistic effects with SPS and Cz-BPDB. The 1000 rmp-PSC bath's potential significantly dropped (−0.15 to −0.23 V) after the addition of Cz-BPDB at 3000 s, and polarization occurred. Additionally, the potential slightly increased (from −0.25 to −0.23 V) after the addition of SPS to the 1000

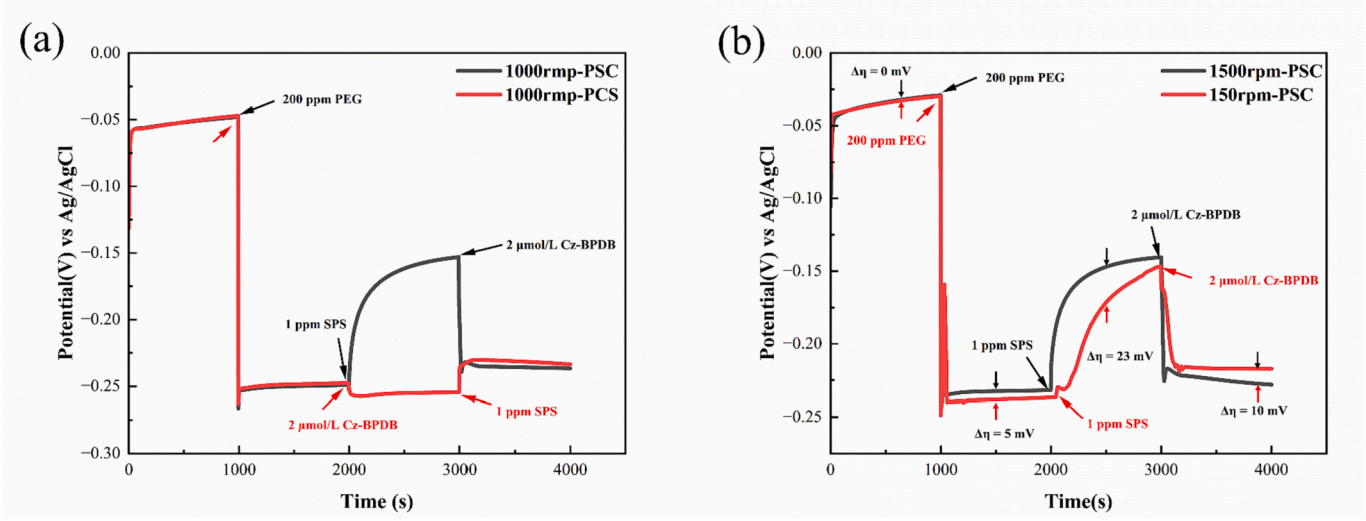


Fig. 14. (a) The GMs plot of different addition sequences. (b) The GMs plot of 1500rmp and 150rmp rotational speeds.

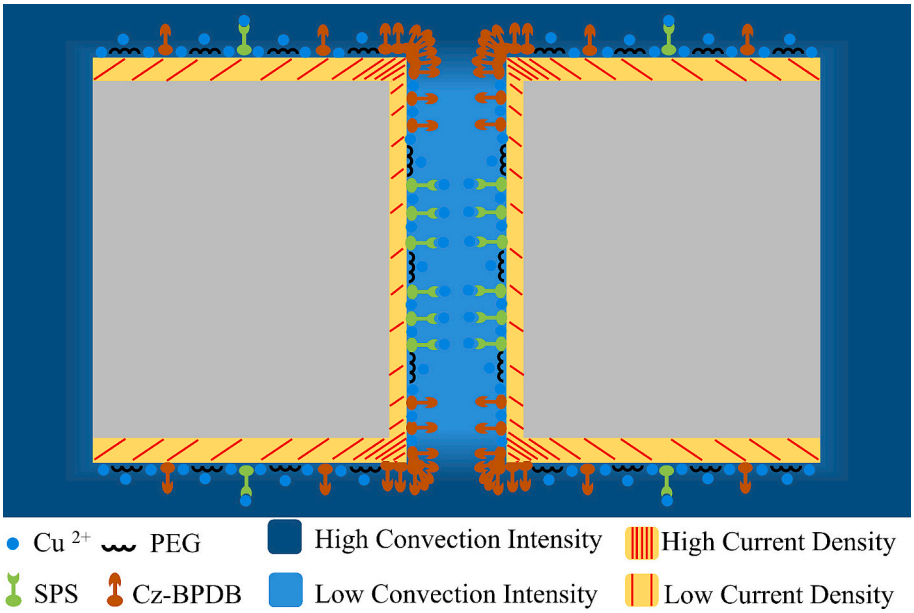


Fig. 15. The additives mechanism diagram for through-hole electroplating.

rmp-PCS bath. It was clear from this that SPS and Cz-BPDB had an antagonistic effect.

As shown in Fig. 14(b), the PSC electroplating solutions were subjected to GMs tests under different rotational speeds of 1500rmp and 150rmp, respectively. The convection velocity was strong at high rmp. The smaller potential difference (Δη) between high and low rotational speeds indicated that the distribution of additives on through hole was less affected by convection velocity and more uniformly distributed inside and outside through hole. According to a previous study, the convection velocity inside through hole was lower than the convection velocity at the orifice [58]. Simultaneously, the current density at the through-hole orifice was high while the current density inside through hole was low [59].

As shown in Fig. 14(b), the Δη went from 0 to 5 mV after the addition of PEG at 1000 s, which indicated the small effect of convection intensity on the adsorption and distribution of PEG. After the addition of SPS at 2000 s, the Δη changed dramatically and was about 23 mV, which indicated the large effect of convection intensity on the adsorption and distribution of SPS. Therefore, the stronger the SPS depolarization phenomenon, the larger number of SPS adsorption in the high convection environment of 1500 rmp. After adding Cz-BPDB at 3000 s, the Δη was 10 mV and became larger with time, which indicated the large effect of convection intensity on the adsorption and distribution of Cz-BPDB. Under the principle of mutual attraction of anisotropic charges, the stronger the polarization of Cz-BPDB in the high convection environment of 1500 rmp, the more Cz-BPDB is adsorbed to the region of high current density.

Based on the above discussion and previous studies [39,60], a reasonable mechanism was introduced. As shown in Fig. 15, the red line represented the current density, and the higher density of the line represented the higher current density. And then, the blue represented the convective intensity, while the dark blue represented the high convective intensity. PEG was more uniformly distributed inside and outside the hole, while the main distribution of the leveler Cz-BPDB was positively correlated with the charge density (i.e., Cz-BPDB was mainly distributed at the orifice). Because PEG and more Cz-BPDB adsorbed to the orifice, the SPS lacked favorable conditions for adsorption at the orifice and adsorbed more to the interior of the holes under the guidance of high convective intensity to promote the reduction and deposition of copper ions inside the holes. It clearly explained the experimental phenomenon of increasing TP values in Fig. 9. Thus, a more reasonable through-hole filling mechanism was proposed for through-hole electroplating with Cz-BPDB used for the leveler.

4. Conclusions

In conclusion, the synthesized carbazole derivatives were used as levelers for copper electroplating. Initially, Cz-BPDB was screened from five carbazole derivatives as the most effective leveler by electrochemical and computational studies. Hereafter, various formulations were designed to investigate the performance of Cz-BPDB. In Bath 5 with 0.5 ppm Cz-BPDB, the TP value of through-hole electroplating reached 104.77 %. Meanwhile, adding Cz-BPDB, the electroplating surface became smooth and flat from roughness as seen by SEM. And after, the dominant crystallographic orientation of the electroplating layer was (111). Further, a series of works were carried out to investigate the mechanism of action of Cz-BPDB. CA tests showed that the nucleation pattern of the VPS + Cz-BPDB solution conformed to the instantaneous nucleation model. The active sites of chemisorption and physisorption of Cz-BPDB were analyzed and localized to the carbazole and pyridine structures of Cz-BPDB in the ALIE plot and the ESP plot. Additionally, the GMs test results illustrated the preferential adsorption sites of the three additives in the electroplating through holes. Eventually, a reasonable mechanism for through-hole electroplating is proposed, which helps to promote the copper electroplating leveler to be further investigated and developed.

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Declaration of competing interest

We declare that we have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.surfcoat.2023.129526.

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